QP CODE: 25901238

Valuation Scheme for MGU-UGP (HONOURS) EXAMINATION, MARCH 2026

FOURTH Semester - MG4SECECT204 -

SKILL ENHANCEMENT COURSE-MULTIMEDIA ELECTRONICS

(2024 ADMISSION ONWARDS)

Duration:1.5Hrs **Maximum Marks:50**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.
*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Part A

Multiple Choice Questions Answer All Questions Each question carries 1 mark.

1.C, K/CO1

2.B. U/CO2

3.B. U/CO3

4.B K/CO1

5.A. U/CO2

6.B. K/CO1

7.C. K/CO2

8.C.. U/CO4

9.A. U/CO2

10.A U/CO4

11.B. K/CO1

12.B. U/CO2

13.A. K/CO3

14.B. U/CO1

15.A. U/CO4

16.C. U/CO1

17.A. U/CO2

18.B U/CO3

19.C. K/CO2

20.A. U/CO4

21.B. U/CO1

22.C. K/CO2

23.C. K/CO4

24.B. U/CO1

25.A. U/CO2

26.B. U/CO1

27.D. U/CO2

28.A. U/CO3

29.B. K/CO1

30.A. U/CO4

31.C. K/CO1

32.A. U/CO2

33.A. K/CO3

34.A. U/CO1

35.C. K/CO2

(1 * 35 = 35)

Part B

Short Answer Questions Answer any 3 Questions Each question carries 5 marks

36.. 1. Definition of Amplitude (1 Mark) **K/CO1**

- Correct definition: Amplitude is the **maximum displacement** of a vibrating particle from its mean position.

2. Explanation of Amplitude (1 Mark)

- Relation to loudness (loud/soft sound).
- Higher amplitude → louder sound; lower amplitude → softer sound.

3. Example for Amplitude (1 Mark)

- Example:
  - Loudspeaker at high volume → large amplitude
  - Whisper → small amplitude

4. Definition of Frequency (1 Mark)

- Correct definition: Frequency is the **number of vibrations per second**, measured in **Hertz (Hz)**.

5. Explanation with Example (1 Mark)

- Relation to pitch (high/low sound).
- Example:
  - Female voice → high frequency (high pitch)
  - Male voice → low frequency (low pitch)

37. 1. Basic Introduction of HDMI (0.5 Mark) **U/CO2**

- HDMI (High Definition Multimedia Interface) is a digital interface used to transmit **audio and video signals**.

2. Basic Introduction of DVI (0.5 Mark)

- DVI (Digital Visual Interface) is used to transmit **video signals only**.

3. Key Differences (3 Marks)

(Any **6 valid comparison points** – 0.5 mark each)

- **Signal Type:**
  HDMI → Audio + Video
  DVI → Video only
- **Audio Support:**
  HDMI → Supports audio
  DVI → No audio support
- **Connector Size:**
  HDMI → Smaller, compact
  DVI → Larger connector
- **Usage:**
  HDMI → TVs, laptops, home theatre
  DVI → Monitors, older PCs
- **Compatibility:**
  HDMI ↔ DVI compatible using adapters (video only)
- **Resolution Support:**
  HDMI → Supports higher resolutions (4K, etc.)
  DVI → Limited compared to HDMI
- **Additional Features:**
  HDMI → Supports CEC, Ethernet (in newer versions)
  DVI → No extra feature

38. 1. Definition of Tone Control (1 Mark) **U/CO3**

- Tone control circuit is used to **adjust the frequency response** of an audio signal by controlling bass and treble.

2. Explanation of Bass Control (1 Mark)

- Bass refers to **low-frequency sounds**.
- Circuit boosts/attenuates low frequencies to enhance depth of sound.

3. Explanation of Treble Control (1 Mark)

- Treble refers to **high-frequency sounds**.
- Circuit boosts/attenuates high frequencies to improve clarity.

4. Working Principle / Circuit Concept (1.5 Marks)

- Uses **RC (Resistor-Capacitor) networks**.
- Capacitors allow high frequencies and block low frequencies.
- By adjusting controls (potentiometers), bass and treble levels are varied.

5. Example / Application (0.5 Mark)
- Used in audio amplifiers, music systems, equalizer

39. 1. Definition of Bitmap Image (1.5 Marks) **U/CO1**
- Bitmap image is a **raster image made up of pixels (tiny dots)** arranged in a grid.
- Each pixel contains color information.

2. Explanation / Key Feature (0.5 Mark)
- Image quality depends on **resolution (number of pixels)**.
- Enlarging the image causes **pixelation**.

3. Advantages (1.5 Marks)
(Any **3 points** – 0.5 mark each)
- High detail and realistic images
- Suitable for photographs
- Supports complex color variations

4. Disadvantages (1.5 Marks)
(Any **3 points** – 0.5 mark each)
- Large file size
- Loss of quality when scaled (pixelation)
- Not suitable for logos/graphics requiring scaling

40. 1. Definition of AV Tools (1 Mark) **U/CO4**
- AV (Audio-Visual) tools are equipment and software used to **record, edit, and present audio and video content**.

2. Role in Video Production (1 Mark)
- Cameras and video tools are used to **capture scenes, visuals, and actions**.

3. Role in Audio Production (1 Mark)
- Microphones and audio tools are used to **record dialogue, background music, and sound effects**.

4. Role in Editing and Post-Production (1.5 Marks)
- Editing software is used to **cut, arrange, add effects, color correction, and synchronize audio with video**.

5. Example / Application (0.5 Mark)
- Used in film making, documentaries, advertisements, and digital media production.

(5 * 3 = 15)